

LLM Explainability

Dataset: South German Credit (UCI Machine Learning Repository, dataset #573)
Model under study: Logistic Regression | LLM tested: gpt-5.6-sol

1. Research Question

The specific test question asked of the LLM in every condition was:
“Explain why the trained logistic-regression model produced this prediction for this applicant. Base your explanation only on the supplied information and explicitly identify any limitations.”

2. Dataset

South German Credit (UCI dataset #573, the corrected/updated version curated by Ulrike Grömping), downloaded directly from the UCI repository at runtime so the pipeline is fully reproducible.

Property	Value
Applicants (rows)	1,000
Raw predictor features	20 (17 categorical, 3 numeric: duration, amount, age)
Target	Credit risk, recoded so 1 = bad credit risk, 0 = good credit risk
Transformed feature count	54 (after standard-scaling numeric features and one-hot encoding categorical features, dropping the first category as reference)

#	Condition	Evidence given to the LLM	Exact explanation possible?
1	customer_prediction_only	Applicant's raw data + the model's prediction only. No information about the algorithm at all.	No
2	textual_description	+ a generic textbook description of logistic regression and feature definitions. No fitted numbers.	No
3	training_code_only	+ the actual scikit-learn training / preprocessing code. Shows how the model was built, but not its learned weights.	No
4	learned_parameters	+ the fitted intercept, all 54 coefficients, and this applicant's exact transformed feature values. No code.	Yes
5	full_structured_package	All of the above combined: description + code + parameters + precomputed contributions.	Yes

3. Test Cases

Applicant ID	True label	Predicted probability (bad credit)	Predicted class
SGC0340	good credit (0)	0.0033	0 (good credit)
SGC0159	bad credit (1)	0.9597	1 (bad credit)
SGC0349	good credit (0)	0.5005	1 (bad credit)

4. Applicant SGC0159

Condition 1 - customer_prediction_only

Prompt : only the applicant's raw fields and the model's predicted probability/class-nothing about the algorithm.

LLM Answer

The supplied model output reports a bad-credit probability of 0.9597154965. Because this exceeds the 0.50 classification threshold, the model assigned class 1, labeled bad_credit. However, the supplied information does not include the fitted intercept, feature coefficients, encoding or transformation details, or applicant-specific feature contributions. Therefore, it is not possible to determine which applicant features most strongly moved the model's bad-credit log-odds or to independently reconstruct the logit, probability, and class. The prediction is a model estimate, not confirmation that the applicant will have bad credit.

Condition 2 - textual_description

Prompt : + a generic explanation of how logistic regression works, plus definitions of all 20 features. Explicitly states that fitted numbers are not supplied.

LLM Answer

The reported bad-credit probability is 0.9597154965. Because this exceeds the 0.50 classification threshold, the model assigns class 1 (bad_credit). This is a model estimate, not confirmation that the applicant will have a bad credit outcome. A feature-level explanation cannot be supported: the fitted intercept, coefficients, transformed applicant values, and applicant-specific contributions were not supplied, so none of the applicant's features can be identified as increasing or decreasing the model's bad-credit log-odds.

Condition 3 - training_code_only

Prompt : + the actual scikit-learn pipeline used to train the model (excerpt below). The fitted intercept and coefficients are deliberately not included.

LLM Answer

The supplied probability corresponds to a bad-credit logit of approximately 3.170671. Because the bad-credit probability of 0.9597154965 exceeds the 0.50 classification threshold, the model assigned class 1 (bad_credit). However, the supplied information does not support identifying which applicant features most strongly moved the prediction or their directions. Feature-level effects cannot be reconstructed without the fitted coefficients, intercept, numeric scaling statistics, and fitted one-hot category ordering. This is a model prediction, not confirmation that the applicant will have bad credit.

Condition 4 - learned_parameters

Prompt : + the fitted intercept and all 54 coefficients, and this applicant's transformed value for every one of them (excerpt below). No training code.

LLM Answer

Using the supplied intercept, coefficients, and transformed applicant values, the reconstructed logit is 3.1706700158 and the corresponding bad-credit probability is 0.9597154965. This exceeds the 0.50 threshold, so the reconstructed class is 1 (bad credit risk). The strongest applicant-specific increases in bad-credit log-odds are `foreign_worker=no` (+0.9246), `installment_rate=installment_rate_below_20_percent` (+0.8178), `duration_standardized` (+0.6496), `property=real_estate` (+0.6072), and `job=skilled_employee_or_official` (+0.5538). Countervailing contributions include `other_installment_plans=none` (-0.4805), `purpose=retraining` (-0.4704), `employment_duration=employed_at_least_7_years` (-0.2910), `status=checking_balance_below_0_DM` (-0.2317), and `people_liable=zero_to_two_people_liable` (-0.2105). Overall, the positive contributions outweighed the negative contributions and the intercept, producing the class-1 prediction. These are fitted model associations, not causal effects, and the prediction is not confirmation of a future outcome.

Condition 5 - full_structured_package

Prompt : the textual description + training code + learned parameters + precomputed contribution table, all combined.

LLM Answer

explanation_possible: True | top-5 recall: 1.00 | direction accuracy: 1.00 | hallucinated features: 0

Starting from the intercept of -0.2895 and summing all supplied applicant-specific contributions gives a logit of 3.1706700156. The strongest contributions by absolute magnitude were `foreign_worker=no` (+0.9246), `installment_rate=installment_rate_below_20_percent` (+0.8178), `duration_standardized` (+0.6496), `property=real_estate` (+0.6072), and `job=skilled_employee_or_official` (+0.5538); each increased the model's bad-credit log-odds. Other supplied features included both increasing and decreasing contributions. Applying the sigmoid function produces a bad-credit probability of 0.9597154965. Because this exceeds the 0.50 threshold, the reconstructed class is 1, labeled `bad_credit`. These are fitted model associations and applicant-specific score contributions, not claims that the features caused the prediction outcome or that bad credit will certainly occur.